

Cramer's rule for coordinates of a row

MathlibAnnex.Matrix.det_smul_vecMul_nonsingInv_eq_updateRowDet

theorem

Expresses a row in the basis of the rows of an invertible square matrix through row-replacement determinants.

Statement

Let A be an n -by- n matrix over a field, with $\det A \neq 0$, and let r be a row of length n . If $A[j \leftarrow r]$ denotes replacement of row j by r , then

$$(\det A)(rA^{-1})_j = \det(A[j \leftarrow r]) \quad (1 \leq j \leq n).$$

Assumptions

The scalar system is an arbitrary field, $n \geq 0$, and r is any row. Nonvanishing of $\det A$ is the exact condition ensuring that the matrix inverse is an inverse.

Conclusion

The coordinate row $c = rA^{-1}$ satisfies $cA = r$, and each coordinate is $c_j = \det(A[j \leftarrow r]) / \det A$. If $n = 0$, the equality is between the two empty coordinate families.

The row convention is essential: the unknown coordinates multiply A on the left. The source obtains this formula by applying the column form of Cramer's rule to the transpose; replacing a column of the transpose replaces the corresponding row before transposition.

Proof route

Apply Cramer's rule to the transpose, then identify each transposed replacement determinant.

▼ Proof steps

1. Write $c = rA^{-1}$. Since $\det A \neq 0$, the inverse exists and

$$cA = r \iff A^T c^T = r^T.$$

Apply the column form of Cramer's rule to the system on the right. Its coefficient matrix satisfies $\det(A^T) = \det A \neq 0$, so it gives

$$(\det A)c_j = \det((A^T)[\text{column } j \leftarrow r^T]).$$

2. The replaced-column matrix is exactly the transpose of the replaced-row matrix:

$$\begin{aligned}(A^T)[\text{column } j \leftarrow r^T] &= (A[j \leftarrow r])^T, \\ (\det A)(rA^{-1})_j &= \det((A[j \leftarrow r])^T) \\ &= \det(A[j \leftarrow r]).\end{aligned}$$

Dividing by $\det A \neq 0$ gives the stated coordinate formula. For $n = 0$ there is no coordinate j to check.

Main citations

- Exact declaration and its source — `MathlibAnnex.Matrix.det_smul_vecMul_nonsingInv_eq_updateRowDet`
- Cramer's rule in the transpose row convention — `Matrix.det_smul_inv_vecMul_eq_cramer_transpose`
- Entries of the transposed Cramer vector — `Matrix.cramer_transpose_apply`

Lean source signature (exact)

```
theorem det_smul_vecMul_nonsingInv_eq_updateRowDet
  {n : ℕ} {K : Type v} [Field K]
  (A : _root_.Matrix (Fin n) (Fin n) K) (r : Fin n → K)
  (hA : A.det ≠ 0) :
  A.det • (r v * A⁻¹) = fun j => (A.updateRow j r).det
```

In the source	Mathematical meaning
<code>{n : ℕ} {K : Type v} [Field K]</code>	An arbitrary field and an integer $n \geq 0$.
<code>(A : root.Matrix (Fin n) (Fin n) K) (r : Fin n → K)</code>	The square matrix A and row vector r of length n .
<code>(hA : A.det ≠ 0)</code>	The hypothesis $\det A \neq 0$, ensuring that A^{-1} is its ordinary matrix inverse.
<code>r v * A⁻¹</code>	The row vector rA^{-1} , rather than matrix times a column.

In the source	Mathematical meaning
<code>fun j => (A.updateRow j r).det</code>	The coordinate family $j \mapsto \det(A[j \leftarrow r])$; row j is replaced and the other rows remain.
<code>A.det • (r v * A⁻¹) = fun j => (A.updateRow j r).det</code>	For every row position j , $(\det A)(rA^{-1})_j = \det(A[j \leftarrow r])$. For $n = 0$ both coordinate families are empty.

▼ Exact surrounding binder context (separate excerpt)

```
noncomputable section

open scoped Matrix
```

```
namespace MathlibAnnex
namespace Matrix

universe u v
```

▼ **Exact content identity**

Declaration: `MathlibAnnex.Matrix.det_smul_vecMul_nonsingInv_eq_updateRowDet`

Accepted content SHA-256: 5659e9ecb00331a1bc04662532630b0fe758fa8181059313bc07568e406a91dd

Accepted source guide SHA-256: 095d5a0faa50d2de6a7f4eb7f83600df7d366888ba9f8f6f290e4a22f7351bb3

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73